

Hessian Geometry, Local Rates, and Acceleration of Sinkhorn

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Abstract

This note isolates one conditional-expectation operator that controls three aspects of entropic optimal transport: dual curvature, the local convergence of Sinkhorn's algorithm, and semi-dual conditioning. At the optimal coupling π_ϵ^* , write $T_\epsilon := T_{\pi_\epsilon^*}$. The positive full-dual curvature operator is $\epsilon^{-1} \begin{pmatrix} I & T_\epsilon \\ T_\epsilon^* & I \end{pmatrix}$, the Jacobian of one complete Sinkhorn cycle is $T_\epsilon^* T_\epsilon$, and exact elimination of one potential gives the semi-dual Hessian $\epsilon^{-1}(I - T_\epsilon^* T_\epsilon)$. This common structure yields the local rate, strict-gap criteria, small- ϵ asymptotics, and explicit parameters for extrapolation and momentum. It also explains why variable projection and quasi-Newton methods are natural alternatives to long scaling iterations.

1 Dual Hessian and conditional expectation

The correct geometry for the dual potentials is not an unweighted Euclidean space: it is the product of the two marginal L^2 spaces. In this geometry, the Hessian has an especially simple block form.

Entropic primal and dual. Let $\mathcal{X}$ and $\mathcal{Y}$ be standard Borel spaces, let $\alpha \in \mathcal{P}(\mathcal{X})$ and $\beta \in \mathcal{P}(\mathcal{Y})$, and let $c : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R} \cup \{+\infty\}$ be measurable. The KL-regularized transport value is

$$\mathcal{L}_c^\epsilon(\alpha, \beta) := \inf_{\pi \in \mathcal{U}(\alpha, \beta)} \left\{ \int c \, d\pi + \epsilon \, \text{KL}(\pi \parallel \alpha \otimes \beta) \right\}. \quad (1)$$

This is the entropy-regularized transport problem introduced computationally by Cuturi [3]. Its concave dual objective is

$$\mathcal{D}_\epsilon(f, g) := \int_{\mathcal{X}} f \, d\alpha + \int_{\mathcal{Y}} g \, d\beta - \epsilon \int_{\mathcal{X} \times \mathcal{Y}} \left[\exp \left(\frac{f(x) + g(y) - c(x, y)}{\epsilon} \right) - 1 \right] d\alpha(x) d\beta(y) \quad (2)$$

and $\mathcal{L}_c^\epsilon(\alpha, \beta) = \sup_{f, g} \mathcal{D}_\epsilon(f, g)$ under the standard duality assumptions. For any potentials for which the integral is finite, denote their Gibbs measure by

$$d\pi_{f, g}(x, y) = \exp \left(\frac{f(x) + g(y) - c(x, y)}{\epsilon} \right) d\alpha(x) d\beta(y) \quad (3)$$

and write $(\pi_{f, g})_1, (\pi_{f, g})_2$ for its marginals. The first variation is

$$D\mathcal{D}_\epsilon(f, g)[h, k] = \int_{\mathcal{X}} h \, d(\alpha - (\pi_{f, g})_1) + \int_{\mathcal{Y}} k \, d(\beta - (\pi_{f, g})_2). \quad (4)$$

Consequently, $\pi_{f, g} \in \mathcal{U}(\alpha, \beta)$ exactly when (f, g) is stationary. Since $\mathcal{D}_\epsilon$ is concave, every such pair is a dual maximizer. Assume henceforth that a maximizer (f^*, g^*) exists, and set

$$\pi_\epsilon^* := \pi_{f^*, g^*} \in \mathcal{U}(\alpha, \beta). \quad (5)$$

The gauge transformation $(f, g) \mapsto (f + s, g - s)$ leaves (2) unchanged. We therefore equip the potentials with the quotient Hilbert space

$$\mathcal{H}_q := \frac{L^2(\alpha) \oplus L^2(\beta)}{\text{span}\{(\mathbf{1}, -\mathbf{1})\}}, \quad \|(h, k)\|_{\mathcal{H}_q}^2 := \inf_{s \in \mathbb{R}} \left\{ \|h + s\mathbf{1}\|_\alpha^2 + \|k - s\mathbf{1}\|_\beta^2 \right\}. \quad (6)$$

Each class has a unique representative orthogonal to the gauge direction, characterized by $\int h \, d\alpha = \int k \, d\beta$. Notice that this condition does not center both potentials: the common-constant direction $(\mathbf{1}, \mathbf{1})$ remains in the quotient.

We also use the centered spaces

$$L_0^2(\alpha) := \left\{ h : \int h \, d\alpha = 0 \right\}, \quad L_0^2(\beta) := \left\{ k : \int k \, d\beta = 0 \right\},$$

endowed with their usual weighted norms. We write

$$\text{cent}_\alpha h := h - \int h \, d\alpha, \quad \text{cent}_\beta k := k - \int k \, d\beta \quad (7)$$

for the orthogonal centering projectors. The product $L_0^2(\alpha) \oplus L_0^2(\beta)$ isolates the nonconstant singular modes, but it has codimension two and is not a representative of the quotient (6).

Coupling operator. The operator governing the local theory is meaningful for every coupling, independently of entropic optimality.

Definition 1.1 (Conditional operator and maximal correlation). For $\pi \in \mathcal{U}(\alpha, \beta)$, disintegrate π with respect to each marginal and define

$$T_\pi : L^2(\beta) \longrightarrow L^2(\alpha), \quad T_\pi k(x) := \mathbb{E}_\pi[k(Y) \mid X = x], \quad (8)$$

$$T_\pi^* : L^2(\alpha) \longrightarrow L^2(\beta), \quad T_\pi^* h(y) := \mathbb{E}_\pi[h(X) \mid Y = y]. \quad (9)$$

Its centered maximal-correlation coefficient is

$$\begin{aligned} \sigma(\pi) &:= \left\| T_\pi : L_0^2(\beta) \rightarrow L_0^2(\alpha) \right\|_{\text{op}} \\ &= \sup_{\substack{h \in L_0^2(\alpha), \, k \in L_0^2(\beta) \\ \|h\|_\alpha = \|k\|_\beta = 1}} \int_{\mathcal{X} \times \mathcal{Y}} h(x)k(y) \, d\pi(x, y). \end{aligned} \quad (10)$$

For the entropic optimizer (5), set

$$T_\epsilon := T_{\pi_\epsilon^*}, \quad \sigma_\epsilon := \sigma(\pi_\epsilon^*). \quad (11)$$

The adjoint notation is literal: $\langle h, T_\pi k \rangle_\alpha = \langle T_\pi^* h, k \rangle_\beta$. Conditional Jensen gives $\|T_\pi\|_{\text{op}} \leq 1$, while $T_\pi \mathbf{1} = \mathbf{1}$ and $T_\pi^* \mathbf{1} = \mathbf{1}$. Hence T_π and T_π^* preserve centered functions and $\sigma(\pi) \in [0, 1]$. The spectral gap of $T_\epsilon^* T_\epsilon$ on $L_0^2(\beta)$ is $1 - \sigma_\epsilon^2$.

Proposition 1.2 (Dual Hessian at arbitrary and optimal potentials). *For arbitrary potentials (f, g) and directions $(h, k), (h', k')$ for which the expressions are finite,*

$$-\epsilon D^2 \mathcal{D}_\epsilon(f, g)[(h, k), (h', k')] = \int_{\mathcal{X} \times \mathcal{Y}} (h(x) + k(y))(h'(x) + k'(y)) \, d\pi_{f, g}(x, y). \quad (12)$$

At an optimal pair (f^, g^*) , the positive curvature operator on $L^2(\alpha) \oplus L^2(\beta)$ is*

$$H_{\text{dual}} = -\nabla^2 \mathcal{D}_\epsilon(f^*, g^*) = \frac{1}{\epsilon} \begin{pmatrix} I & T_{\pi_\epsilon^*} \\ T_{\pi_\epsilon^*}^* & I \end{pmatrix}. \quad (13)$$

The Hessian descends to the quotient $\mathcal{H}_q$ in (6), where

$$\frac{1 - \sigma_\epsilon}{\epsilon} I \preceq H_{\text{dual}} \preceq \frac{2}{\epsilon} I. \quad (14)$$

Consequently, if $\sigma_\epsilon < 1$, its spectral condition number is

$$\kappa_{\text{dual}} = \frac{2}{1 - \sigma_\epsilon}. \quad (15)$$

If $\sigma_\epsilon = 1$, the quotient Hessian is not coercive and this condition number is infinite.

Proof. Twice differentiating the exponential term in $-\mathcal{D}_\epsilon$ gives (12). At the optimum, $\pi_{f^\star, g^\star} = \pi_\epsilon^\star$ has marginals (α, β) . The two diagonal terms in (12) are therefore the $L^2(\alpha)$ and $L^2(\beta)$ inner products, while the cross term is $\langle h, T_{\pi_\epsilon^\star} k \rangle_\alpha$. This proves (13). On a centered singular pair $T_\epsilon v = su$, $T_\epsilon^* u = sv$, the two vectors $(u, \pm v)$ have eigenvalues $(1 \pm s)/\epsilon$. A representative orthogonal to the gauge decomposes uniquely as

$$(h, k) = (h_0, k_0) + m(\mathbf{1}, \mathbf{1}), \quad h_0 \in L_0^2(\alpha), \quad k_0 \in L_0^2(\beta).$$

The two summands are orthogonal and invariant under the Hessian. The first has spectral interval $[(1 - \sigma_\epsilon)/\epsilon, (1 + \sigma_\epsilon)/\epsilon]$, while the second has eigenvalue $2/\epsilon$. Since $1 + \sigma_\epsilon \leq 2$, this proves (14) and (15). $\square$

Thus optimality is not needed for the second-variation identity (12). It is needed for the clean conditional-expectation form (13): away from stationarity, $\pi_{f, g}$ is generally not a coupling, the diagonal blocks are multiplication by its marginal densities relative to α and β , and the off-diagonal block is an unnormalized integral operator rather than T_π from Definition 1.1.

For discrete marginals $\alpha = \sum_i a_i \delta_{x_i}$ and $\beta = \sum_j b_j \delta_{y_j}$, with optimal coupling matrix $P_\epsilon^\star$, the same operators read

$$T_{\pi_\epsilon^\star} k = D_a^{-1} P_\epsilon^\star k, \quad T_{\pi_\epsilon^\star}^* h = D_b^{-1} (P_\epsilon^\star)^\top h,$$

where $D_a = \text{diag}(a)$ and $D_b = \text{diag}(b)$. These formulas are adjoint in the a - and b -weighted inner products, not in the unweighted Euclidean ones.

2 Sinkhorn Jacobian and local rate

A full Sinkhorn cycle is a nonlinear map on one potential. Its linearization at the fixed point is exactly the two-step conditional expectation $T_\epsilon^* T_\epsilon$, so the local convergence question reduces to a Markov spectral gap.

Soft transforms and the cycle map. Exact block maximization of $\mathcal{D}_\epsilon$ is written with the continuous soft c -transforms

$$g^{\bar{c}, \epsilon}(x) := -\epsilon \log \int_{\mathcal{Y}} \exp \left(\frac{g(y) - c(x, y)}{\epsilon} \right) d\beta(y), \quad (16)$$

$$f^{c, \epsilon}(y) := -\epsilon \log \int_{\mathcal{X}} \exp \left(\frac{f(x) - c(x, y)}{\epsilon} \right) d\alpha(x). \quad (17)$$

One complete cycle on the target potential is therefore

$$\mathcal{S}_\epsilon(g) := (g^{\bar{c}, \epsilon})^{c, \epsilon}, \quad g^{(\ell+1)} = \mathcal{S}_\epsilon(g^{(\ell)}). \quad (18)$$

The opposite ordering acts on $L^2(\alpha)$ and has the same nonzero spectrum.

Proposition 2.1 (Jacobian and exact local Sinkhorn rate). *Assume that the two soft transforms are twice differentiable near the optimum. Then*

$$D[g \mapsto g^{\bar{c}, \epsilon}](g^*) = -T_\epsilon, \quad D[f \mapsto f^{c, \epsilon}](f^*) = -T_\epsilon^*, \quad D\mathcal{S}_\epsilon(g^*) = T_\epsilon^* T_\epsilon. \quad (19)$$

Consequently, for the centered error $e^{(\ell)} = \text{cent}_\beta(g^{(\ell)} - g^*)$ of any Sinkhorn sequence converging to g^* modulo constants,

$$e^{(\ell+1)} = T_\epsilon^* T_\epsilon e^{(\ell)} + o(\|e^{(\ell)}\|_\beta), \quad \limsup_{\ell \rightarrow \infty} \frac{\|e^{(\ell+1)}\|_\beta}{\|e^{(\ell)}\|_\beta} \leq \sigma_\epsilon^2. \quad (20)$$

If σ_ϵ^2 is an isolated eigenvalue and the asymptotic error has a nonzero projection onto its eigenspace, the ratio converges to σ_ϵ^2 .

Proof. Differentiating the log-partition maps (16)–(17) at the optimal Gibbs coupling gives the first two identities in (19); the chain rule gives the third. Taylor expansion proves (20). $\square$

The reverse cycle $f \mapsto (f^{c, \epsilon})^{\bar{c}, \epsilon}$ has Jacobian $T_\epsilon T_\epsilon^*$ on $L_0^2(\alpha)$. Since $T_\epsilon^* T_\epsilon$ and $T_\epsilon T_\epsilon^*$ have the same nonzero spectrum, either orientation has the same local rate. In finite spaces, this recovers the classical second-singular-value formula for matrix scaling [12, 13].

When is the rate strictly smaller than one? Positivity of the Gibbs kernel is enough in finite dimension but not on an arbitrary infinite space: centered approximate modes may retain correlation arbitrarily close to one. In the finite or compact continuous setting, strictness follows directly from the same Birkhoff–Hilbert contraction theorem that gives global linear convergence of Sinkhorn. The only additional step is to differentiate that global estimate at the fixed point.

Let $k_\epsilon(x, y) := e^{-c(x, y)/\epsilon}$. For compact supports and continuous c , define the four-point oscillation

$$\Delta_c := \sup_{x, x', y, y'} [c(x, y') + c(x', y) - c(x, y) - c(x', y')], \quad (21)$$

where the supremum is over the product of the two supports. This quantity is invariant under $c(x, y) \mapsto c(x, y) + \xi(x) + \zeta(y)$ and is exactly the projective diameter of the Gibbs operator, up to temperature:

$$\sup_{x, x', y, y'} \log \frac{k_\epsilon(x, y) k_\epsilon(x', y')}{k_\epsilon(x, y') k_\epsilon(x', y)} = \frac{\Delta_c}{\epsilon}. \quad (22)$$

Proposition 2.2 (Birkhoff–Hilbert bound on the local rate). *Assume either finite spaces with a positive Gibbs kernel, or compact supports with continuous finite c . Set*

$$\Lambda_\epsilon := \tanh\left(\frac{\Delta_c}{4\epsilon}\right) < 1. \quad (23)$$

Each Gibbs half-step is Λ_ϵ -contractive in Hilbert’s projective metric. Consequently, a complete Sinkhorn cycle is globally Λ_ϵ^2 -contractive in Hilbert’s projective metric, and its local L^2 factor satisfies

$$\sigma_\epsilon^2 \leq \Lambda_\epsilon^2, \quad \text{hence} \quad \sigma_\epsilon \leq \Lambda_\epsilon < 1. \quad (24)$$

Proof. Equation (22) and the Birkhoff–Hopf theorem give the half-step estimate (23); multiplication and inversion in the scaling updates are projective isometries, so a full cycle has factor Λ_ϵ^2 [7]. In logarithmic coordinates, Hilbert’s metric is the variation seminorm. Differentiating the full-cycle contraction at its fixed point and using Proposition 2.1 therefore gives

$$\text{osc}(T_\epsilon^* T_\epsilon k) \leq \Lambda_\epsilon^2 \text{osc}(k).$$

In finite dimension this applies directly to a top centered eigenvector of $T_\epsilon^* T_\epsilon$. In the compact continuous case, the conditional operator is compact and that eigenvector is continuous. Its eigenvalue is σ_ϵ^2 , which proves (24); the case $\sigma_\epsilon = 0$ is immediate. $\square$

A cost-oscillation estimate modulo separable terms is only a coarser way to bound the same Birkhoff coefficient: for every ξ, ζ ,

$$\Delta_c \leq 2 \operatorname{ess\,osc}_{\alpha \otimes \beta} (c(x, y) - \xi(x) - \zeta(y)).$$

Thus it introduces no distinct contraction mechanism and need not be proved separately. On unbounded spaces Δ_c is generally infinite, so the universal Birkhoff estimate becomes vacuous. Cost growth must then be balanced against marginal tails. Conforti, Durmus, and Greco prove quantitative contraction for quadratic costs under asymptotically positive log-concavity profiles [2]; Eckstein obtains weighted Hilbert contraction under compatible cost-growth and tail assumptions [6]. These are useful input-level criteria, but their norms should not be identified automatically with the exact L^2 gap $1 - \sigma_\epsilon^2$.

Why a linear small- ϵ gap is natural. The conditional-variance identity becomes useful specifically for understanding the small-regularization regime. The law of total variance gives

$$1 - \sigma_\epsilon^2 = \inf_{\substack{k \in L_0^2(\beta) \\ \|k\|_\beta = 1}} \mathbb{E}_{\pi_\epsilon^*} [\operatorname{Var}(k(Y) \mid X)]. \quad (25)$$

Indeed, the right-hand side is the smallest Rayleigh quotient of $I - T_\epsilon^* T_\epsilon$ on $L_0^2(\beta)$. In a smooth connected Monge regime, the conditional law of Y given $X = x$ has width $O(\sqrt{\epsilon})$ around the hard transport map. Consequently, the conditional variance in (25) is typically of order ϵ , leading to $\sigma_\epsilon^2 = 1 - C\epsilon + o(\epsilon)$.

A precise conditional statement is as follows. Assume that $\mathcal{X}$ and $\mathcal{Y}$ are smooth connected d -dimensional manifolds, that the marginals have positive smooth densities, and that hard transport is induced by a smooth diffeomorphism $\mathbf{M} : \mathcal{X} \rightarrow \mathcal{Y}$. Let g_0^* be the hard target potential and set

$$H(x) := D_{yy}^2 (c(x, \cdot) - g_0^*)(\mathbf{M}(x)), \quad A(y) := H(\mathbf{M}^{-1}(y))^{-1}. \quad (26)$$

Suppose that H is uniformly positive and that the uniform Laplace expansion of the entropic conditionals permits passage to the first nonzero Rayleigh quotient, for instance through Mosco convergence of the rescaled forms and compact convergence of their resolvents. Define

$$\lambda_1(A, \beta) := \inf_{\substack{\int k \, d\beta = 0 \\ \int k^2 \, d\beta = 1}} \int_{\mathcal{Y}} \langle A(y) \nabla k(y), \nabla k(y) \rangle \, d\beta(y). \quad (27)$$

Uniform ellipticity and a Poincaré inequality for β ensure that $\lambda_1(A, \beta) > 0$.

Proposition 2.3 (Smooth small-regularization law). *Under the preceding assumptions,*

$$\sigma_\epsilon^2 = 1 - \epsilon \lambda_1(A, \beta) + o(\epsilon), \quad \sigma_\epsilon = 1 - \frac{\epsilon}{2} \lambda_1(A, \beta) + o(\epsilon). \quad (28)$$

Proof. For the optimal coupling π_ϵ^* , the uniform Laplace expansion of $Y \mid X = x$ gives $\operatorname{Cov}(Y \mid X = x) = \epsilon H(x)^{-1} + o(\epsilon)$. Hence, for smooth k ,

$$\frac{1}{\epsilon} \mathbb{E}_{\pi_\epsilon^*} [\operatorname{Var}(k(Y) \mid X)] \longrightarrow \int_{\mathcal{X}} \langle H(x)^{-1} \nabla k(\mathbf{M}(x)), \nabla k(\mathbf{M}(x)) \rangle \, d\alpha(x).$$

Since $\mathbf{M}_\# \alpha = \beta$, the right-hand side is the form in (27). Spectral convergence permits passage to the first Rayleigh quotient in (25), giving the expansion of σ_ϵ^2 . The expansion of σ_ϵ follows by taking the square root. $\square$

Example 2.4 (Exact Gaussian benchmark). For $\alpha = \beta = \mathcal{N}(0, 1)$ and $c(x, y) = \frac{1}{2}(x - y)^2$, the optimal coupling is Gaussian with correlation

$$r_\epsilon = \frac{\sqrt{\epsilon^2 + 4} - \epsilon}{2}. \quad (29)$$

Hermite polynomials diagonalize T_ϵ , so $\sigma_\epsilon = r_\epsilon$. Thus $\sigma_\epsilon = 1 - \epsilon/2 + O(\epsilon^2)$ and $\sigma_\epsilon^2 = 1 - \epsilon + O(\epsilon^2)$. This is Proposition 2.3 with $A = 1$ and $\lambda_1(A, \beta) = 1$.

Local L^2 versus global Hilbert contraction. For $\Delta_c > 0$,

$$1 - \Lambda_\epsilon^2 \sim 4e^{-\Delta_c/(2\epsilon)} \quad (\epsilon \downarrow 0). \quad (30)$$

This is exponentially more pessimistic than the smooth local law $1 - \sigma_\epsilon^2 \sim \epsilon \lambda_1(A, \beta)$. There is no contradiction. Hilbert's projective metric gives a global guarantee over the full positive cone and uses no smoothness of the iterates. The L^2 result is local to the optimal coupling and exploits the regular diffusive structure of the dominant modes. Chizat, Delalande, and Vaškevičius similarly obtain polynomial rather than universal exponential dependences under continuous regularity and log-concavity assumptions [1].

3 Extrapolation and momentum

Once the local spectrum is reduced to $\text{spec}(T_\epsilon^* T_\epsilon) \subset [0, \sigma_\epsilon^2]$, standard polynomial acceleration applies. Scalar extrapolation and momentum are two instances of the same stationary two-step iteration.

Two-parameter acceleration. For $\omega > 0$ and $\vartheta \geq 0$, consider

$$g^{(\ell+1)} = g^{(\ell)} + \omega(\mathcal{S}_\epsilon(g^{(\ell)}) - g^{(\ell)}) + \vartheta(g^{(\ell)} - g^{(\ell-1)}). \quad (31)$$

The choice $\vartheta = 0$ gives scalar extrapolation, while $(\omega, \vartheta) = (1, 0)$ is ordinary Sinkhorn. Locally, (31) is the heavy-ball iteration applied to $I - T_\epsilon^* T_\epsilon$ [16]. On a mode with $T_\epsilon^* T_\epsilon k = \lambda k$, write $a := 1 - \lambda \in [1 - \sigma_\epsilon^2, 1]$; the scalar error satisfies

$$z_{\ell+1} = (1 - \omega a + \vartheta)z_\ell - \vartheta z_{\ell-1}. \quad (32)$$

Proposition 3.1 (Optimal extrapolation and momentum parameters). *Assume $\sigma_\epsilon < 1$.*

1. Among iterations (31) with $\vartheta = 0$, the minimax choice and its worst local factor are

$$\omega_{\text{ext}} = \frac{2}{2 - \sigma_\epsilon^2}, \quad r_{\text{ext}} = \frac{\sigma_\epsilon^2}{2 - \sigma_\epsilon^2}. \quad (33)$$

2. Among stationary two-step iterations (31) with $\omega > 0$ and $\vartheta \geq 0$, the minimax parameters are

$$\omega_{\text{mom}} = \frac{4}{(1 + \sqrt{1 - \sigma_\epsilon^2})^2}, \quad \vartheta_{\text{mom}} = \left(\frac{1 - \sqrt{1 - \sigma_\epsilon^2}}{1 + \sqrt{1 - \sigma_\epsilon^2}} \right)^2. \quad (34)$$

Their worst local asymptotic factor is

$$r_{\text{mom}} = \frac{1 - \sqrt{1 - \sigma_\epsilon^2}}{1 + \sqrt{1 - \sigma_\epsilon^2}}. \quad (35)$$

Thus extrapolation doubles the small gap to first order, whereas momentum changes its scale from $1 - \sigma_\epsilon^2$ to $\sqrt{1 - \sigma_\epsilon^2}$.

Proof. If $\vartheta = 0$, the mode factor is $1 - \omega a$. Balancing its values at $a = 1 - \sigma_\epsilon^2$ and $a = 1$ gives (33).

For the two-step problem, the case $\sigma_\epsilon = 0$ is immediate. Assume $0 < \sigma_\epsilon < 1$ and suppose that both roots of $z^2 - (1 - \omega a + \vartheta)z + \vartheta$ have modulus at most $r < 1$ for every $a \in [1 - \sigma_\epsilon^2, 1]$. Applying the elementary quadratic unit-disk criterion after rescaling the roots by r gives

$$0 \leq \vartheta \leq r^2, \quad -r - \frac{\vartheta}{r} \leq 1 - \omega a + \vartheta \leq r + \frac{\vartheta}{r}.$$

Applying the upper inequality at $a = 1 - \sigma_\epsilon^2$ and the lower one at $a = 1$, then using $\vartheta/r \leq r$, yields the necessary condition

$$(1 - r)^2 \leq (1 - \sigma_\epsilon^2)(1 + r)^2.$$

Hence $r \geq (1 - \sqrt{1 - \sigma_\epsilon^2})/(1 + \sqrt{1 - \sigma_\epsilon^2})$. For this lower-bound value, (34) gives $\vartheta = r^2$ and makes $1 - \omega a + \vartheta$ range from $2r$ to $-2r$. Both roots therefore have modulus at most r , proving optimality. $\square$

A one-parameter momentum curve. For a fixed gap, minimax tuning equioscillates the endpoint modes, which imposes the linear relation

$$2(1 + \vartheta) = \omega(2 - \sigma_\epsilon^2), \quad \text{or equivalently} \quad \vartheta = \frac{2 - \sigma_\epsilon^2}{2}\omega - 1. \quad (36)$$

At the optimum, a more convenient one-parameter representation is

$$\vartheta = r^2, \quad \omega = (1 + r)^2, \quad 1 - \sigma_\epsilon^2 = \left(\frac{1 - r}{1 + r}\right)^2, \quad 0 \leq r < 1. \quad (37)$$

Thus selecting r is equivalent to selecting an estimated gap and automatically couples the two algorithmic parameters. This reduces parameter search to one dimension. It does not remove the need for safeguards: an overestimate of $1 - \sigma_\epsilon^2$ can produce parameters that are unstable for the true spectral interval.

For $1 - \sigma_\epsilon^2 \ll 1$, scalar extrapolation changes σ_ϵ^2 into $1 - 2(1 - \sigma_\epsilon^2) + O((1 - \sigma_\epsilon^2)^2)$ and therefore preserves the scaling of the iteration count. In contrast, $r_{\text{mom}} = 1 - 2\sqrt{1 - \sigma_\epsilon^2} + O(1 - \sigma_\epsilon^2)$.

Over-relaxing the two block updates separately gives the nonlinear SOR variant studied in [18, 13]. Its optimal local parameter is

$$\omega_{\text{SOR}} = \frac{2}{1 + \sqrt{1 - \sigma_\epsilon^2}},$$

and it has the same factor (35). Block SOR and the outer momentum step (31) are different algorithms, despite their identical optimal local rate. Their global convergence requires safeguards; the displayed parameters are local spectral prescriptions, not unconditional global guarantees.

Estimating the gap from iterates. The optimal parameters depend on an unknown local quantity. During an ordinary Sinkhorn run, set

$$d^{(\ell)} := \text{cent}_\beta(g^{(\ell)} - g^{(\ell-1)}).$$

Near the fixed point, $d^{(\ell+1)} = T_\epsilon^* T_\epsilon d^{(\ell)} + o(\|d^{(\ell)}\|)$, so the Rayleigh quotient

$$\hat{r}_\ell := \frac{\langle d^{(\ell)}, d^{(\ell+1)} \rangle_\beta}{\|d^{(\ell)}\|_\beta^2} \quad (38)$$

converges generically to σ_ϵ^2 when a dominant mode is present. The norm ratio $\|d^{(\ell+1)}\|_\beta / \|d^{(\ell)}\|_\beta$ is a simpler, slightly noisier alternative. In practice one clips $\hat{r}_\ell$ to $[0, 1)$, uses $1 - \hat{r}_\ell$ as an estimate of $1 - \sigma_\epsilon^2$ in the formulas above, and activates acceleration only after the estimate has stabilized. This also explains why extrapolation based on early iterates can be unreliable: the asymptotic spectral mode has not yet emerged.

4 Semi-dual variable projection

Eliminating one potential exactly converts the same operator $T_\epsilon^* T_\epsilon$ from an iteration Jacobian into a Hessian correction. This is the variable-projection mechanism: the poorly conditioned block coupling is removed by a Schur complement rather than handled by alternating block maximization.

This elimination is a special case of the variable-projection method of Golub and Pereyra [8, 9]. In general, if one block is optimized exactly (or minimized for the negative objective) and the positive Hessian at a stationary point is $H = \begin{pmatrix} H_{11} & H_{12} \\ H_{21} & H_{22} \end{pmatrix}$ with $H_{11} \succ 0$, then the reduced Hessian is

$$H_{\text{red}} = H_{22} - H_{21} H_{11}^{-1} H_{12}. \quad (39)$$

Its variational characterization gives, with extremal spectral values in infinite dimension, $\lambda_{\min}(H) \leq \lambda_{\min}(H_{\text{red}})$ and $\lambda_{\max}(H_{\text{red}}) \leq \lambda_{\max}(H)$, so exact projection cannot worsen the local spectral condition number after the gauge has been removed. General variable-projection theory does not, however, quantify this gain without additional information on the blocks. Here both *diagonal* blocks of the normalized full-dual Hessian are the identity, while the off-diagonal blocks are T_ϵ and T_ϵ^* . This special structure makes the reduced operator $I - T_\epsilon^* T_\epsilon$ and its conditioning explicitly computable from σ_ϵ .

Reduced objective and Hessian. For each g , the exact source-potential maximizer is $g^{\bar{c}, \epsilon}$ from (16). The resulting concave semi-dual is

$$\mathcal{E}_\epsilon(g) := \mathcal{D}_\epsilon(g^{\bar{c}, \epsilon}, g) = \int_{\mathcal{X}} g^{\bar{c}, \epsilon} d\alpha + \int_{\mathcal{Y}} g d\beta. \quad (40)$$

Let

$$d\pi_g(x, y) := \exp\left(\frac{g^{\bar{c}, \epsilon}(x) + g(y) - c(x, y)}{\epsilon}\right) d\alpha(x) d\beta(y), \quad (41)$$

and denote its second marginal by β_g . Its first marginal is exactly α .

Proposition 4.1 (Semi-dual derivatives and Schur complement). *For bounded directions k, k' on $\mathcal{Y}$,*

$$D\mathcal{E}_\epsilon(g)[k] = \int_{\mathcal{Y}} k d(\beta - \beta_g), \quad (42)$$

$$D^2\mathcal{E}_\epsilon(g)[k, k'] = -\frac{1}{\epsilon} \int_{\mathcal{X}} \text{Cov}_{\pi_g(\cdot|x)}(k(Y), k'(Y)) d\alpha(x). \quad (43)$$

At $g = g^*$, the positive Riesz curvature operator on $L^2(\beta)$ is

$$H_{\text{sd}} := -\nabla^2 \mathcal{E}_\epsilon(g^*) = \frac{1}{\epsilon} (I - T_\epsilon^* T_\epsilon). \quad (44)$$

The identity extends from bounded directions to $L^2(\beta)$ by density. It is exactly the Schur complement of the f block in (13).

Proof. The envelope theorem gives (42). Differentiating the conditional Gibbs law in (41) gives the conditional covariance formula (43). At the optimum,

$$\int_{\mathcal{X}} \text{Cov}(k, k' \mid X) d\alpha = \langle k, k' \rangle_{\beta} - \langle T_{\epsilon} k, T_{\epsilon} k' \rangle_{\alpha} = \langle k, (I - T_{\epsilon}^* T_{\epsilon}) k' \rangle_{\beta},$$

which proves (44). Finally, the Schur complement of I in $\begin{pmatrix} I & T_{\epsilon} \\ T_{\epsilon}^* & I \end{pmatrix}$ is $I - T_{\epsilon}^* T_{\epsilon}$, which is (39) in the present geometry. $\square$

Conditioning and first-order rates. The spectral gap that controls Sinkhorn is exactly the strong concavity of $\mathcal{E}_{\epsilon}$, or equivalently the strong convexity of $-\mathcal{E}_{\epsilon}$.

Proposition 4.2 (Semi-dual conditioning and optimization rates). *On $L_0^2(\beta)$,*

$$\frac{1 - \sigma_{\epsilon}^2}{\epsilon} I \preceq H_{\text{sd}} \preceq \frac{1}{\epsilon} I, \quad \kappa_{\text{sd}} \leq \frac{1}{1 - \sigma_{\epsilon}^2}. \quad (45)$$

Gradient ascent on the local quadratic model of $\mathcal{E}_{\epsilon}$, with optimal constant step $2\epsilon/(2 - \sigma_{\epsilon}^2)$, has factor

$$r_{\text{GD}} \leq \frac{\sigma_{\epsilon}^2}{2 - \sigma_{\epsilon}^2}. \quad (46)$$

Heavy-ball or Chebyshev acceleration has factor

$$r_{\text{HB}} \leq \frac{1 - \sqrt{1 - \sigma_{\epsilon}^2}}{1 + \sqrt{1 - \sigma_{\epsilon}^2}}. \quad (47)$$

Proof. The spectrum of $T_{\epsilon}^* T_{\epsilon}$ on centered functions lies in $[0, \sigma_{\epsilon}^2]$, so the spectrum of (44) lies in $[(1 - \sigma_{\epsilon}^2)/\epsilon, 1/\epsilon]$. This proves the curvature and condition-number bounds. The standard minimax polynomials on an interval give (46) and (47). $\square$

Compared with the quotient full-dual condition number (15), the displayed semi-dual upper bound is smaller by the factor $2(1 + \sigma_{\epsilon})$. This is the quantitative gain furnished here by variable projection.

Quasi-Newton methods. In a finite discretization, $-\mathcal{E}_{\epsilon}$ is a smooth unconstrained convex problem after fixing its additive gauge. BFGS attempts to learn the inverse of $I - T_{\epsilon}^* T_{\epsilon}$ from gradient differences, precisely the operator responsible for the slow modes. Under the standard local smoothness and positive-definiteness assumptions, full BFGS is locally superlinear; limited-memory BFGS trades that guarantee for lower memory and often strong practical performance [14].

This strategy has a substantial OT history. Cuturi and Peyré used smooth dual and semi-dual formulations with L-BFGS for variational Wasserstein problems and developed the general regularized semi-dual viewpoint [4, 5]; stochastic first-order optimization of semi-duals was developed by Genevay et al. [10]. Khamlich, Romor, and Rozza use L-BFGS as the optimization core of a modern entropy-regularized semi-discrete solver, together with truncation, multilevel continuation, and fast geometric queries [11]. Qiu, Yin, and Wang also report and analyze L-BFGS on the fully discrete entropic dual as an alternative to long Sinkhorn runs [17].

These methods do not make the small- ϵ conditioning disappear: the inverse Hessian still grows like $1/(1 - \sigma_{\epsilon}^2)$. Their advantage is that curvature information is learned and reused. Each quasi-Newton step is more expensive than one scaling update, so comparisons should count kernel evaluations or matrix-vector products rather than iterations. The semi-dual is especially attractive when one block can be eliminated analytically or integrated accurately, as in semi-discrete transport.

Remark 4.3 (Extrapolated Sinkhorn versus semi-dual gradient ascent). These methods are not identical globally. For the coupling π_g in (41), let $r_g := d\beta_g/d\beta$. The scaling identities give

$$r_g = \exp\left(\frac{g - \mathcal{S}_\epsilon(g)}{\epsilon}\right), \quad \nabla \mathcal{E}_\epsilon(g) = 1 - r_g, \quad \mathcal{S}_\epsilon(g) - g = -\epsilon \log r_g. \quad (48)$$

Thus a Sinkhorn cycle uses the logarithmic marginal defect, whereas semi-dual gradient ascent uses the additive defect. They agree only to first order near $r_g = 1$, where

$$\mathcal{S}_\epsilon(g) - g = \epsilon \nabla \mathcal{E}_\epsilon(g) + o(\|g - g^*\|). \quad (49)$$

Consequently, relaxed Sinkhorn with parameter ω is locally semi-dual gradient ascent with step $\epsilon\omega$, and (31) is locally its heavy-ball counterpart with momentum ϑ . This explains the agreement between (33)–(35) and (46)–(47), without claiming a global identity. Nesterov acceleration has the same square-root dependence on $1 - \sigma_\epsilon^2$ [15], although its standard worst-case constant differs from the stationary heavy-ball factor.

Operator summary. All local statements in the note follow from one contraction:

Object	Operator at the optimum	Consequence
Full dual Hessian	$\epsilon^{-1} \begin{pmatrix} I & T_\epsilon \\ T_\epsilon^* & I \end{pmatrix}$	coupled conditioning
Sinkhorn full-cycle Jacobian	$T_\epsilon^* T_\epsilon$	local factor σ_ϵ^2
Semi-dual Hessian	$\epsilon^{-1}(I - T_\epsilon^* T_\epsilon)$	curvature $(1 - \sigma_\epsilon^2)/\epsilon$
Accelerated local polynomial	$I - T_\epsilon^* T_\epsilon$	rate governed by $\sqrt{1 - \sigma_\epsilon^2}$

Thus the slow convergence of Sinkhorn and the ill-conditioning of the semi-dual are complementary descriptions of the same near-deterministic correlation modes. Variable projection removes one coupled block; extrapolation changes the first-order polynomial; momentum and quasi-Newton methods approximate the inverse curvature more aggressively.

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